

The Complete S_4 Chebotarev–Frobenius Permutation Atlas for the Root Scheme of $x^4 - x - 1$

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Abstract

The integral quartic $f(x) = x^4 - x - 1$ defines an intrinsic prime-indexed family of finite dynamical systems: arithmetic Frobenius $F_p(\alpha) = \alpha^p$ permutes the four geometric roots at every good prime. We prove that $\text{disc}(f) = -283$ and $\text{Gal}(f/\mathbb{Q}) = S_4$, identify factor degrees with primitive cycle lengths, and close the fixed-point and finite-determinant identities on every good fiber. All five S_4 cycle types are exhibited. The final round adds their Chebotarev densities, the non-étale boundary at $p = 283$, and the exact fiberwise Koopman classification. The construction is a source-arithmetic candidate, not a cross-prime target determinant. The universal finite-permutation trace/determinant mechanism remains owned by the prior workspace package HCS-C12A; our owner is the complete $x^4 - x - 1$ -specific atlas and its executable convention lock.

1 Quartic arithmetic and the S_4 proof

Put

$$f(x) = x^4 - x - 1, \quad \mathcal{X} = \text{Spec } \mathbb{Z}[x]/(f). \quad (1)$$

For a rational prime p , write X_p for the geometric roots of the reduction of f in $\overline{\mathbb{F}}_p$. We use *arithmetic Frobenius*

$$F_p : X_p \longrightarrow X_p, \quad \alpha \longmapsto \alpha^p. \quad (2)$$

Geometric Frobenius is F_p^{-1} . The two conventions have identical cycle lengths, fixed counts, and determinants; this paper never conflates the maps, only their common cycle partition.

Theorem 1 (Arithmetic and Galois closure). *The discriminant of f is -283 . The polynomial is irreducible over $\mathbb{Q}$ and its splitting field L has*

$$\text{Gal}(L/\mathbb{Q}) \cong S_4.$$

Every prime $p \neq 283$ is unramified and $\mathcal{X}_{\mathbb{F}_p}$ is a finite étale scheme of degree four. If $\lambda_p = (d_1, \dots, d_k)$ is the multiset of degrees of the distinct irreducible factors of $f \bmod p$, then λ_p is exactly the cycle partition of F_p on X_p .

Proof. For $x^4 + qx + r$ the discriminant is $256r^3 - 27q^4$; inserting $q = r = -1$ gives $-256 - 27 = -283$. Modulo 2,

$$f = x^4 + x + 1.$$

It has no root, and the only monic irreducible quadratic over $\mathbb{F}_2$ is $x^2 + x + 1$, whose square is $x^4 + x^2 + 1$. Hence $f \bmod 2$ is irreducible. Gauss’s lemma makes f irreducible over $\mathbb{Q}$, so its Galois group $G \leq S_4$ is transitive. Dedekind’s theorem [1, 2] at the unramified prime 2 supplies a 4-cycle. At $p = 7$,

$$f \equiv (x - 3)(x^3 + 3x^2 + 2x - 2) \pmod{7}, \quad (3)$$

and the cubic factor has no root, so G also contains a 3-cycle. Consequently 12 divides $|G|$, while $|G|$ divides 24. A subgroup of order 12 in S_4 is the index-two sign kernel A_4 , but a 4-cycle is odd; equivalently, the discriminant is not a square. Therefore $G = S_4$.

The discriminant criterion makes precisely $p = 283$ exceptional. At every other prime, the roots are distinct. An irreducible factor of degree d has its roots in one orbit of $\alpha \mapsto \alpha^p$ of length d , and every orbit arises this way [3]. This proves the dictionary. $\square$

Proposition 1 (Five exact witnesses). *Every partition of four occurs among the good fibers. In addition to (3), one has*

$$\begin{array}{lll}
 p = 2 : & f = x^4 + x + 1 & (4), \\
 p = 17 : & f = (x + 2)(x + 5)(x^2 - 7x + 5) & (2 + 1 + 1), \\
 p = 71 : & f = (x^2 + 15x - 20)(x^2 - 15x + 32) & (2 + 2), \\
 p = 83 : & f = (x + 3)(x + 7)(x + 14)(x - 24) & (1 + 1 + 1 + 1),
 \end{array}$$

where every equality is in the corresponding residue field and the final parenthesis is the Frobenius cycle partition.

Proof. Multiplication gives f modulo the displayed prime. The linear factors are explicit; the remaining quadratic or cubic factors have no residue-field root. The mod-2 case was proved in Theorem 1. $\square$

References

- [1] J.-P. Serre, *Topics in Galois Theory*, second edition, A K Peters, 2008.
- [2] J. Neukirch, *Algebraic Number Theory*, Springer, 1999.
- [3] R. Lidl and H. Niederreiter, *Finite Fields*, second edition, Cambridge University Press, 1997.